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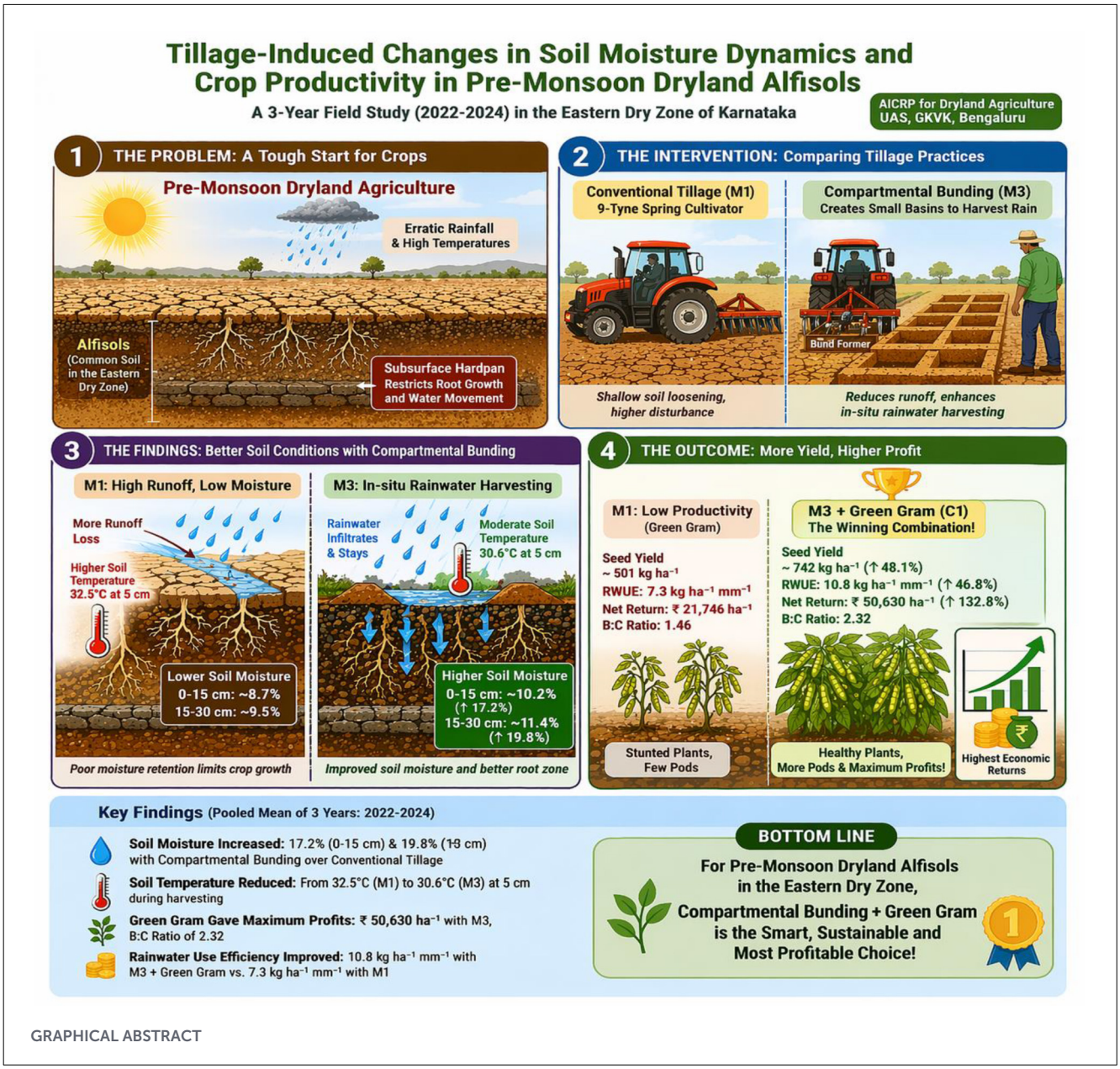
Soil moisture dynamics, thermal regulation and crop yield responses to tillage practices in dryland Alfisols

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A 3-year field experiment (2022–2024) was conducted during the pre-monsoon season (March–May) on Alfisols at the All India Coordinated Research Project for Dryland Agriculture, University of Agricultural Sciences, Bengaluru, India, to evaluate the effects of moisture conservation practices on soil moisture dynamics, crop productivity, and profitability. The experiment was laid out in a split-plot design with three moisture conservation practices: conventional tillage (M_1), duck foot cultivator (M_2), and compartmental bunding (M_3) as main plots, and three crops: green gram (*Vigna radiata* L.) var. KKM-3, field bean (*Dolichos lablab* L.) var. HA-4, and sesame (*Sesamum indicum* L.) var. GKVKS-1 as subplots. Soil moisture content at both 0–15 and 15–30 cm depths was significantly ($P \leq 0.05$) influenced by moisture conservation practices, with compartmental bunding consistently maintaining greater soil moisture and lower soil temperature than conventional tillage. Pooled over 3 years, compartmental bunding significantly increased seed yield (894 kg ha^{-1}), green gram equivalent yield (GGEY) (1,276 kg ha^{-1}), and rainwater use efficiency (RWUE) (2.58 kg ha-mm^{-1}) by 23.0%, 23.5%, and 24.6%, respectively, over conventional tillage. It also improved net returns and benefit-cost ratio by 34.2% and 15.7%, respectively. Among crops, field bean produced significantly greater seed yield (1,099 kg ha^{-1}) and rainwater use efficiency (2.78 kg ha-mm^{-1}) than sesame, with increases of 153.2% and 74.8%, respectively, while green gram generated the highest net returns (61,194 ha^{-1}) and benefit-cost ratio (2.85). Heatmap analysis identified the treatment combination $M3C_2$ (compartmental bunding + field bean) as the best-performing system, recording 997 kg ha^{-1} seed yield, 1,298 kg ha^{-1} GGEY, 2.68 kg ha-mm^{-1} RWUE, and a sustainable yield index (SYI) of 0.80. The findings indicate that utilizing compartmental bunding and field beans is a effective strategy for enhancing soil moisture retention, increasing productivity, improving water-use efficiency, and boosting profitability in pre-monsoon rainfed Alfisols.

KEYWORDS
compartmental bunding, conventional tillage, duck foot cultivator, soil moisture, soil temperature



GRAPHICAL ABSTRACT

1 Introduction

Agricultural productivity in dryland ecosystems is fundamentally dependent on the effective capture, retention, and utilization of rainfall, particularly in regions where irrigation facilities are limited or entirely absent. In the semi-arid tropics, especially during the pre-monsoon period (March–May), crop production is frequently constrained by highly erratic rainfall patterns, elevated evaporative demand, and the rapid depletion of soil moisture reserves (Rockström et al., 2003). These constraints have become increasingly severe under changing climatic conditions marked by greater rainfall variability and prolonged dry spells, thereby intensifying the vulnerability of rainfed production systems (Jat et al., 2020; FAO, 2021). In this context, the adoption of efficient *in-situ* moisture conservation practices has emerged as a crucial strategy for sustaining soil water availability, enhancing water-use efficiency, improving crop productivity, and strengthening the resilience of dryland agricultural systems (Cárceles Rodríguez et al., 2022).

In India's dryland agro-ecosystems, crop productivity is largely constrained by the erratic and limited southwest monsoon rainfall, making efficient soil moisture management crucial for sustainable production. Tillage practices strongly influence soil physical properties and soil water dynamics. Appropriate tillage can improve soil aggregation, macroporosity, infiltration, and root growth while reducing runoff and evaporative losses under moisture stress conditions (Thierfelder and Wall, 2009; Lal, 2020). However, the effectiveness of tillage practices varies considerably depending on soil type, cropping system, and climatic conditions, resulting in different responses across agro-ecological regions (Giller et al., 2009; Mărunţelu, 2020). In Alfisols, soil compaction reduces pore space, limits infiltration, and increases

runoff, leading to substantial moisture losses (Lipiec and Hatano, 2003). Subsurface hardpans further impede root proliferation and restrict water movement within the soil profile, thereby intensifying moisture stress conditions. These limitations are aggravated by continuous tillage operations, frequent use of heavy machinery, and livestock trampling, which collectively increase soil penetration resistance, reduce porosity, and promote surface crust formation. Such degradation of soil physical conditions also enhances susceptibility to runoff and soil erosion (Cogo et al., 1983; Gicheru et al., 2004; Shivonje et al., 2005).

Tillage and land configuration practices play a pivotal role in regulating soil water dynamics by influencing infiltration, runoff, evaporation, and root-zone moisture availability. Conventional tillage, involving repeated soil disturbance, often accelerates soil moisture loss through enhanced evaporation and deterioration of soil structure, thereby limiting crop performance under moisture-deficit conditions (Lal, 2020; Busari et al., 2015). In contrast, conservation-oriented tillage practices improve soil physical conditions and facilitate greater rainfall retention within the soil profile (Jat et al., 2014; Adil, 2022; Teng et al., 2024). Among these, the duck foot cultivator has emerged as a promising implement for dryland agriculture due to its ability to loosen the surface soil, improve infiltration, reduce surface crusting, and control weeds with minimal soil inversion. By maintaining better soil aggregation and reducing moisture losses, duck foot cultivation can improve root proliferation and enhance water availability to crops during critical growth stages (Heidari Soltanabadi et al., 2008; Dong et al., 2021; Wang et al., 2023). Such interventions have been shown to enhance soil moisture storage, rainwater use proficiency, and harvest productivity under dryland circumstances (Choudhary et al., 2024; Wani et al., 2007). However, their effectiveness varies across environments, necessitating location-specific evaluation.

Similarly, compartmental bunding is recognized as an effective *in-situ* rainwater harvesting technique in rainfed farming systems. The creation of small compartments across the field reduces runoff velocity, promotes water infiltration, and enhances soil moisture storage within the root zone (Dashrathrao et al., 2023; Kumar et al., 2023). Compartmental bunding substantially improves rainwater use efficiency, reduces soil erosion, and stabilizes crop yields under moisture-limited environments (Jat et al., 2014; Momin et al., 2021). Unlike conventional tillage, subsoiling breaks hardpans without inverting the topsoil, increasing soil permeability and infiltration while refining moistness saving and drought resilience (Xiao et al., 2011). The efficacy of subsoiling rest on on factors such as soil type, climate, crop species, and tillage depth, necessitating site-specific evaluation. Subsoiling can increase subsoil water storage by up to 76.2 mm compared to non-subsoiled conditions, enhanced water storage at depths up to 100 cm highlights its potential for improving crop performance under moisture-limited conditions (Yuan et al., 2025).

Crop productivity in dryland systems is often constrained by low soil moisture and poor nutrient availability. Subsoiling improves soil structure, enhances moisture retention, and increases yield, especially in drought-prone areas (Shen et al., 2021). Soil compaction from machinery, improper tillage, and livestock increases bulk density and reduces porosity, negatively affecting soil health and crop growth (Hakansson and Lipiec, 2000; Hamza and Anderson, 2005). Subsoiling alleviates compaction by reducing bulk density, increasing porosity, and lowering penetration resistance, thereby improving water movement and promoting deeper root growth under moisture stress (Chen and Weil, 2011). Enhanced porosity also supports microbial activity and nutrient cycling. However, the effectiveness of subsoiling depends on soil texture, moisture conditions, and frequency of operation.

The response of crops to moisture conservation practices is largely governed by their rooting behavior, growth duration, and water requirements. Green gram is an important short-duration pulse crop widely cultivated in rainfed regions due to its adaptability and economic value. However, its productivity is highly dependent on adequate moisture availability during flowering and pod development stages, making it responsive to improved soil moisture conservation measures (Sharma et al., 2019; Kumar et al., 2023). Field bean possesses a deeper and more extensive root system, enabling efficient extraction of soil moisture from deeper horizons and better adaptation to intermittent drought conditions.

Consequently, field bean often exhibits improved productivity and water-use efficiency under conservation-based management practices (Dhangada et al., 2017; Duarte et al., 2025). Sesame an important oilseed crop in dryland agriculture, is generally considered drought tolerant; nevertheless, moisture stress during early vegetative growth and reproductive development can significantly affect biomass accumulation and seed yield (Ardeshna et al., 2021; Adil, 2022). Understanding the differential response of these crops to tillage and moisture conservation interventions is therefore essential for developing resilient and profitable dryland production systems.

Although considerable research has documented the benefits of conservation agriculture and moisture conservation practices individually, comparative evaluations of conventional tillage, duck foot cultivation, and compartmental bunding across pulse and oilseed-based cropping systems under pre-monsoon Alfisol conditions remain limited. Addressing this knowledge gap is crucial for identifying appropriate resource-conserving technologies capable of sustaining crop production in moisture-stressed environments. Therefore, the present study was undertaken to evaluate the effects of different tillage and moisture conservation practices on soil moisture retention, soil temperature regulation, crop growth, productivity, rainwater use efficiency, and profitability of green gram, field bean and sesame grown under pre-monsoon rainfed Alfisol.

2 Material and methods

2.1 Site description

The experiment was conducted at the University of Agricultural Sciences, GKVK, Bengaluru, All India Coordinated Research Project for Dryland Agriculture. Located at 13°05′ N latitude and 77°34′ E longitude in Karnataka's Eastern Dry Zone (Zone V), the site is 924 m above mean sea level. The location of the experiment was presented in Figure 1. The Vijayapura series soils, which are typical of the Bengaluru plateau's lateritic region, define

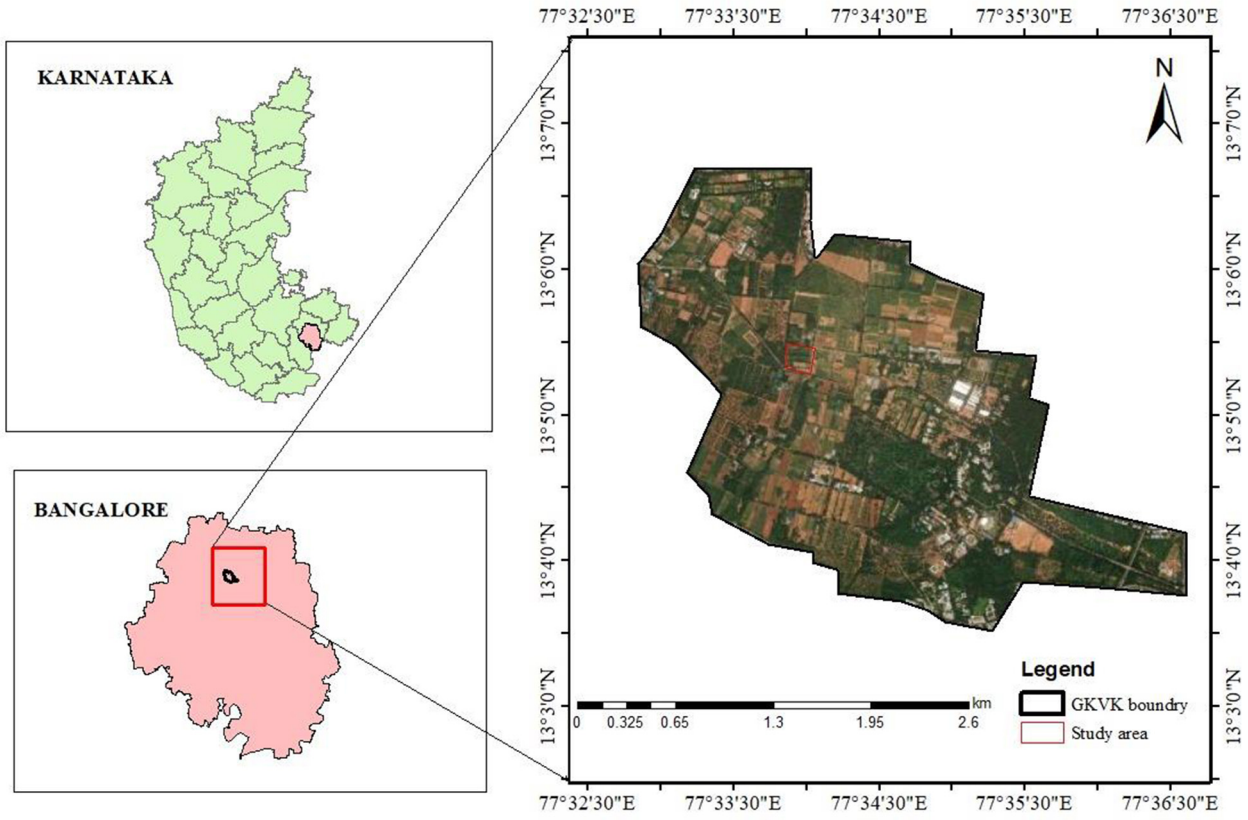


FIGURE 1
Location of the trial area.

the experimental field. These soils are ideal for dryland agricultural research owing to their deep, yellowish-red lateritic red sandy clay loam texture, good drainage, and mild fertility (AICRPDA, 2024). The weather parameters prevailing during the experimental duration for each year (2022 to 2024) was depicted in the Figure 2.

Using Keen's cup method and the process outlined by Piper (1966), the physical characteristics of the soil, including bulk density, particle density, maximum water-holding capacity, and porosity, were ascertained. According to Parr and Bertrand (1960), a double-ring infiltrometer was used to quantify the steady-state infiltration rate (cm h^{-1}). Throughout the crop growth period, the moisture content of the soil was measured every 7 days at two depths (15 and 30 cm). An ML3 Theta Probe was employed for measurements at a depth of 15 cm, while a PR2 Profile Probe was utilized for measurements at a depth of 30 cm. Using a soil thermometer, the temperature of the soil was measured during two crucial stages of crop growth: flowering and harvesting. Table 1 shows the basic physical characteristics of the soil that were examined.

Five randomly selected plants in each plot were used to record crop growth characteristics, which were then averaged. Plant height was measured in centimeters from base to apex, and each plant fully developed trifoliate leaves and branches were tallied. A leaf area meter (cm^2) was used to measure the leaf area, and ground area and leaf area were used to calculate the Leaf Area Index (LAI). Total dry matter per plant was determined by oven-drying plant samples to constant weight and summing their biomass.

Yield and yield-attributing parameters were recorded from sampled plants. The numeral of pods per plant was totaled from fully developed pods, and test weight (g) was determined by weighing a known number of seeds. Grain yield was recorded from harvested produce and expressed as kg ha^{-1}, while stalk yield was estimated from oven-dried biomass and expressed on a hectare basis. Economic analysis was performed by considering input costs and prevailing market prices to compute the cost of farming, gross returns, and net returns (ha^{-1}).

2.2 Experimental design and treatment details

The trial was accompanied during *Pre-kharif* 2022 to 2024 (3 years) at AICRP for Dryland Agriculture, UAS, GKVK, Bengaluru. A split-plot design with three replications. The main plot treatments consisted of three tillage practices M_1: Conventional tillage (nine-tyne spring cultivator), M_2: Duck foot cultivator (five-tyne), and M_3: Compartmental bund former and three cover crops as green manure in subplot treatments C_1: Green gram, C_2: Field bean, and C_3: Sesamum. The nine treatment combinations were replicated thrice across 27 plots. Each experimental unit was laid out according to the treatment combinations under the split-plot arrangement. Recommended practices, including seed treatment, nutrient, irrigation, weed, and plant protection management were uniformly followed for green gram, field bean, and sesame as per

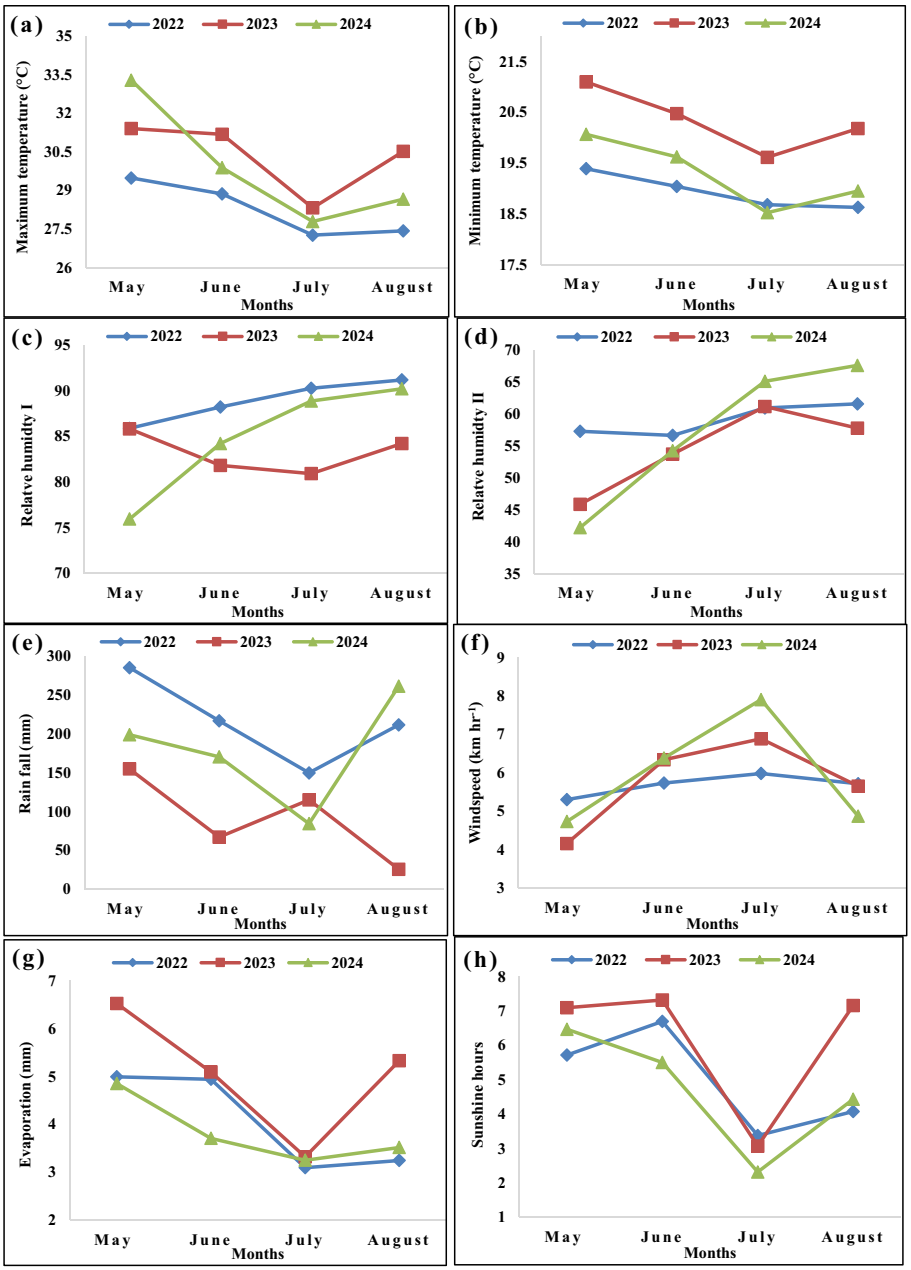


FIGURE 2
Weather parameters prevailing during the cropping period for each year (2022 to 2024). **(a)** Maximum temperature (°C). **(b)** Minimum temperature (°C). **(c)** Relative humidity-I (%). **(d)** Relative humidity-II (%). **(e)** Rainfall (mm). **(f)** Wind speed (km hr^{-1}). **(g)** Evaporation (mm), and **(h)** sunshine hours (hr) of the study area.

regional agronomic recommendations throughout the cropping period. The standard agronomic practices followed and crop variety used details are presented in Table 2.

Conventional tillage (M_1) was carried out using a nine-tyne spring cultivator drawn by a tractor. This implement pulverizes the soil by repeatedly cutting and lifting it, ensuring thorough loosening, breaking up clods, and removing surface weeds. It creates a fine tilth, facilitating good seed-soil contact for uniform germination and root establishment. The duck foot cultivator (M_2), equipped with five tyne blades shaped like a duck's foot, is designed for shallow tillage and inter-cultivation. It slices through the soil horizontally, cutting weeds and loosening the topsoil without inverting it, which helps in moisture conservation and reduces soil compaction. The compartmental bund former (M_3) is an implement used to form small cross ridges or bunds at regular intervals across the field. These bunds divide the land into small compartments, which reduce runoff, enhance *in-situ* rainwater harvesting and improve soil moisture retention by impeding lateral flow of water during rainfall events.

2.3 Soil sampling, performance evaluation and grain yield

Using a screw auger, representative soil samples were taken from each plot at depths of 0 to 15 cm and 15 to 30 cm at 15, 30,

TABLE 1 Soil physical properties of the site.

Particulars	Values	Methods employed
Coarse sand (%)	36.45	International pipet method (Piper, 1966)
Fine sand (%)	33.75	
Silt (%)	8.64	
Clay (%)	21.32	
Soil texture class	Sandy loam	
Infiltration rate (cm h^{-1})	4.29	Double ring infiltrometer method (Parr and Bertrand, 1960)
Maximum water holding capacity (%)	37.43	Keen's cup method (Piper, 1966)
Bulk density (g cc^{-1})	1.48	
Particle density (g cc^{-1})	2.568	
Porosity (%)	55.89	

45, 60, and 75 DAS and at harvest. The physical characteristics of the soil were analyzed for a range of values, as listed below.

2.3.1 Soil physical properties

Soil moisture is detected using the ML3 theta probe (for 15 cm depth) and the ML3 Theta probe soil moisture sensor. It is simple to obtain reliable, precise soil moisture assessments. Simply insert the soil moisture gauge into the soil and it connects to the data logger or display unit within seconds. Soil moisture was measured at a depth of 30 cm using the PR2 profile probe. First, insert an access tube into the soil, insert the PR2 profile probe, and press the HH2 [Read] button to display an instantaneous reading. Then, calibrate the ML3 Theta Probe and PR2 Profile Probe using the gravimetric method (Piper, 1966), and calculate soil moisture content using the formula.

$$\text{Soil moistness content (\%)} = \frac{\text{Wet soil weight (g)} - \text{Dry soil weight (g)}}{\text{Dry soil weight (g)}} \times 100 \quad (1)$$

Soil bulk density and particle density were measured after the harvest of the crop using Keen's cup method (Black, 1965) and expressed in g. cc^{-1}.

$$\text{Porosity (\%)} = (1 - \frac{\rho_b}{\rho_p}) \quad (2)$$

Where, ρ_b is the bulk density and ρ_p is the average particle density.

Soil compaction was determined after sub soiling treatment was imposed, after harvest of the crop in each year, by using a penetrometer at 15, 30, and 45 cm and is expressed in pounds per inch. Soil temperature at 2 stages, the flowering and harvesting stage by soil thermometer.

2.3.2 Performance evaluation of different tillage equipment

Width of cut (m): Average working width of the implement measured across the field.

$$W = \frac{W1 + W2 + W3}{3} \quad (3)$$

Tillage depth (cm): Average depth to which the soil is tilled during operation.

$$D = \frac{D1 + D2 + D3}{3} \quad (4)$$

Speed of operation (ms^{-1}): Distance traveled by the machine per unit time.

$$S = \frac{d}{t} \quad (5)$$

Wheel slip (%): Difference between theoretical and actual travel speed of the tractor expressed in percentage (Jokiniemi et al., 2012).

$$\text{Wheel slip (\%)} = \frac{N_L - N_O}{N_L} \quad (6)$$

For the marked test run, N_L and N_O represent the total number of rotations at load and no load, respectively.

Effective field capacity (ha h^{-1}): Actual rate of area covered by the implement considering total time taken (Pradhan et al., 2016).

$$EFC = \frac{A}{T} \quad (7)$$

Where,

A = area covered (ha), T = total time (h)

Theoretical field capacity (hah^{-1}): Maximum possible rate of field coverage at full width and rated speed without losses (Agricultural Engineering, 2012).

$$TFC = \frac{S \times W}{10} \quad (8)$$

Where,

S = speed (km h^{-1}), W = width (m)

Field capacity (%): Ratio of effective field capacity to theoretical field capacity expressed in percentage (Pradhan et al., 2016).

$$\text{Field capacity} = \frac{\text{Effective field capacity}}{\text{Theoretical field capacity}} \times 100 \quad (9)$$

Draft calculation (kgf): Force required to pull the implement, measured using a dynamometer as the difference between loaded and unloaded conditions (Al-Hamed et al., 2011).

$$\text{Draft} = D_1 - D_2 \quad (10)$$

Where,

D_1 = Dynamometer reading with implement in operation, D_2 = Reading with implement lifted

Fuel consumption (lph): Amount of fuel used by the tractor during operation, determined from the fuel refilled after the test (Mamkagh, 2019).

$$\text{Fuel consumption} = \frac{F}{T} \quad (11)$$

TABLE 2 Standard agronomic practices followed for all three crops.

Particulars	Green gram	Field bean	Sesame
Scientific name and variety	*Vigna radiata* L.) var. KKM-3	*Dolichos lablab* L. var. HA-4	*Sesamum indicum*. L. var. GKVKS-1
Season	May to July	May to August/September	May to August
Seed rate	20 kg ha^{-1}	25 kg ha^{-1}	5 kg ha^{-1}
Spacing (cm)	30 × 10	45 × 15	30 × 10
Sowing depth	3–5 cm	3–5 cm	2–3 cm
Seed treatment	Carbendazim @ 2 g kg^{-1} seed + *Rhizobium* and PSB @ 25 g kg^{-1} seed		Carbendazim @ 2 g kg^{-1} seed + *Azospirillum/Azotobacter* @ 25 g kg^{-1} seed
Fertilizer dose (kg ha^{-1})	12.5: 25: 25 NPK	25:50:25 NPK	37.5:25:25 NPK
Method of fertilizer application	Entire dose applied as basal		Half N and full P & K as basal; remaining N at 30 DAS
Weed management	Pendimethalin @ 1.0 kg a.i. ha^{-1} PE + one hand weeding at 20-25 DAS		
Major insect pest and protection measures	Profenofos spray for pod borer in green gram and field bean, Spray of Deltamethrin 2.8 EC @ 1.5 ml/l for severe infestation of Aphids		
Crop duration	65–70 days	90–105 days	85–95 days
Gross plot	9 m × 9 m = 81 m^2		
Net plot	7.8 m × 8.6 m = 67.08 m^2		

Where,
F = fuel used (liters), T = time of operation (h)

Power requirement (hp): Drawbar horsepower, a measure of the pulling power of an engine by way of tire, wheels, or tracks.

$$\text{Drawbar power (kW)} = \text{Draft (kN)} \times \text{Speed } (ms^{-1}) \quad (12)$$

Conversion of kW power to horsepower is: 1 hp = 0.746 kW.

Rain water use efficacy (RWUE)

$$\text{RWUE (kg/ha} - \text{mm)} = \frac{\text{Crop yield (Y)}}{\text{Seasonal rainfall (mm)}} \quad (13)$$

Where,
Y = Crop yield (kg/ha)
R = Seasonal rainfall (mm)

2.4 Productivity assessment

The system productivity was measured by converting various crop yields into a single equivalent yield namely green gram equivalent yield (GGEY), to enable standardized comparison.

$$\text{Green gram equivalent yield (kg/ha)} = \frac{\text{Yield of a crop} \times \text{Market value of the crop}}{\text{Market value of Green gram}} \quad (14)$$

2.5 Economic evaluation

The economic performance of the farming systems was assessed using key financial indicators, including Net Return (NR) and Benefit-Cost Ratio (BCR). The market prices of inputs and outputs prevailing during the respective production periods were considered.

2.6 Sustainable yield index

Measure of yield stability over time, calculated using mean yield, variability, and maximum yield (Singh et al., 1990).

$$\text{Sustainable Yield Index } (SYI) = \frac{\text{Y} - \sigma}{\text{Y max}} \quad (15)$$

Where,
Y = mean yield
σ = standard deviation
Y max = maximum yield

2.7 Statistical analysis

The analysis of variance (ANOVA) for all assessed characteristics was performed utilizing R software (version 4.5.3) via R Studio (R Core Team, 2020), and R was employed for multivariate analysis and data visualization. Mean comparisons along with grouping letters were determined using Fisher's Least Significant Difference (LSD) test at a significance level of 5% (Gomez and Gomez, 1984).

3 Results and discussion

3.1 Soil moisture content (%)

The boxplot analysis, supported by ANOVA, indicated that soil moisture content was considerably ($p < 0.05$) affected by treatments at both soil depths (0–15 cm and 15–30 cm) across all growth stages (15–90 DAS). Clear differences in median values, variability, and distribution patterns were observed among treatments. At the 0–15 cm depth, treatment M_3 consistently noted

the highest median soil moisture with lower variability, reflecting stable moisture retention. In contrast, M_1 showed the lowest median values with greater dispersion, indicating poor moisture conservation. Treatments C_2 and M_2 exhibited intermediate performance, while C_3 remained in the lower range (Figure 3A). A comparable trend was evident at the 15–30 cm depth, where M_3 maintained superior soil moisture levels across all stages, with relatively uniform distribution. Moisture content at this depth was generally higher and more stable than at the surface, suggesting improved sub-surface water storage. Treatments C_2 and M_2 showed moderate performance, whereas M_1 and C_3 consistently recorded lower values (Figure 3B). The boxplots revealed clear separation among treatments, with minimal overlap between higher- and lower-performing groups, which was further supported by LSD grouping. Across the 3 years (2022–2024), the results consistently showed that M_3 outperformed M_1 and M_2 in conserving soil moisture at both depths. Higher moisture levels were maintained even during relatively dry periods, indicating improved water retention under this treatment.

The superior soil moisture retention observed under M_3 across both depths can be attributed to enhanced soil structural properties, including greater aggregation, pore continuity, and infiltration capacity. These characteristics facilitate better water retention at the surface and increased storage in deeper soil films. Similar improvements in soil moisture dynamics under conservation practices have been reported in recent studies (Jat et al., 2020; Thierfelder and Steward, 2022). Higher moisture availability at the 15–30 cm depth suggests reduced evaporative losses and improved vertical movement of water, which enhances moisture availability within the root zone. This is particularly beneficial under rainfed conditions, where sub-surface moisture plays a critical role in nourishing crop growth during dry spells (Kassam et al., 2019). In contrast, the lower soil moisture recorded under M_1 indicates that conventional tillage may adversely affect soil water conservation by increasing soil disturbance and accelerating evaporation losses. Similar observations have been reported by Busari et al. (2015), emphasizing the negative impact of intensive tillage on soil moisture retention. The intermediate performance of M_2 and C_2 suggests that modified or reduced tillage practices can improve moisture conservation compared to conventional methods, although their effectiveness remains lower than that of M_3. The consistent superiority of M_3 across years further confirms its suitability as a sustainable soil management practice. These findings are in agreement with recent reports showing that practices such as compartmental bunding and conservation tillage improve in-situ water retention and reduce runoff losses (Momin et al., 2021; Jat et al., 2014; Dong et al., 2021). Adopting suitable conservation-based soil management strategies can dramatically increase soil moisture availability, improve water-use efficiency, and boost crop output in changing climates.

3.2 Soil temperature at 5 cm and 10 cm depth

During the flowering stage, at a depth of 5 cm, the highest soil temperature was recorded with the Conventional Tillage method (M_1) at 28.1 °C, followed by the Duck Foot Cultivator (M_2) at 26.7 °C and the Compartmental Bund Former (M_3) at 25.9 °C. Among the crops, Greengram (C_1) exhibited the highest temperature at 27.6 °C, followed by Sesamum (C_3) at 26.8 °C and Field Bean (C_2) at 26.2 °C. Where in case of the second depth (10 cm), M_1 also had the highest temperature at 27.5 °C, followed by M_2 at 26.4 °C and M_3 at 25.7 °C. Similarly, Green Gram (C_1) recorded the highest temperature at 27.6 °C, followed by Field Bean (C_2) and Sesamum (C_3) at 26.0 °C. Moving to the harvesting stage, similar outcomes were attained as in the flowering stage during pooled years (2022–2024).

Conventional tillage (M_1) consistently resulted in the highest soil temperatures, while Compartmental Bunding (M_3) resulted in the lowest temperatures due to better moisture conservation (Table 3). Furthermore, Green Gram (C_1) consistently exhibited higher temperatures across all tillage methods, which bring into line with the conclusions of Wang et al. (2023). The observed variation in soil temperature across different tillage practices and crops in the present study aligns with earlier research findings. Higher soil temperatures under conventional tillage (M_1), which reached up to 33.9 °C at 5 cm depth during harvesting, can be attributed to greater soil exposure and lower surface residue cover, enhancing solar heat absorption. This is consistent with the outcomes of Jat et al. (2011), stated that conventional tillage systems exhibit higher surface soil temperatures compared to conservation practices due to disruption of soil structure and reduced shading. Conversely, compartmental bunding (M_3) effectively moderated soil temperature, recording the lowest values (as low as 29.4 °C at 5 cm), and due to its ability to conserve surface moisture and reduce heat transfer, as noted by Ghosh et al. (2010). Further, the impact of crop type on soil temperature is supported by Vieira et al. (2021), who emphasized that crop canopy development alters the microclimate, with denser canopies typically moderating soil temperature more effectively. However, in the present study, green gram (C_1) exhibited relatively higher soil temperatures, likely due to rapid early growth and higher biomass production, which may have increased respiration and metabolic heat release around the root zone. Boudiar et al. (2022), highlighted that conservation practices, such as bunding and appropriate crop selection, can effectively regulate soil thermal regimes under dryland agriculture (Table 3).

3.3 Growth attributes

The figure highlights the influence of moisture conservation practices and crop types on key plant growth parameters such as plant height, number of leaves plant^{-1}, number of branche plant^{-1}, and leaf area index (LAI) (Figure 4). Among the moisture conservation methods, compartmental bunding (M_3) showed superior performance, recording the highest plant height (103.1 cm), number of branches and leaves plant^{-1} (6.6 and 41.6), and LAI and TDM (3.5 and 0.9 g plant^{-1}), suggesting improved soil moisture retention and favorable microclimatic conditions for crop growth. In contrast, conventional tillage (M_1) resulted in the lowest growth attributes, indicating limited soil moisture conservation. Among crops, Sesamum (C_3) outperformed other crops, with the tallest plants (132.6 cm), the highest number of leaves and branches plant^{-1} (57.5 and 6.0, respectively), maximum LAI and TDM (1.3 and 1.6 g plant^{-1}), reflecting its deeper root

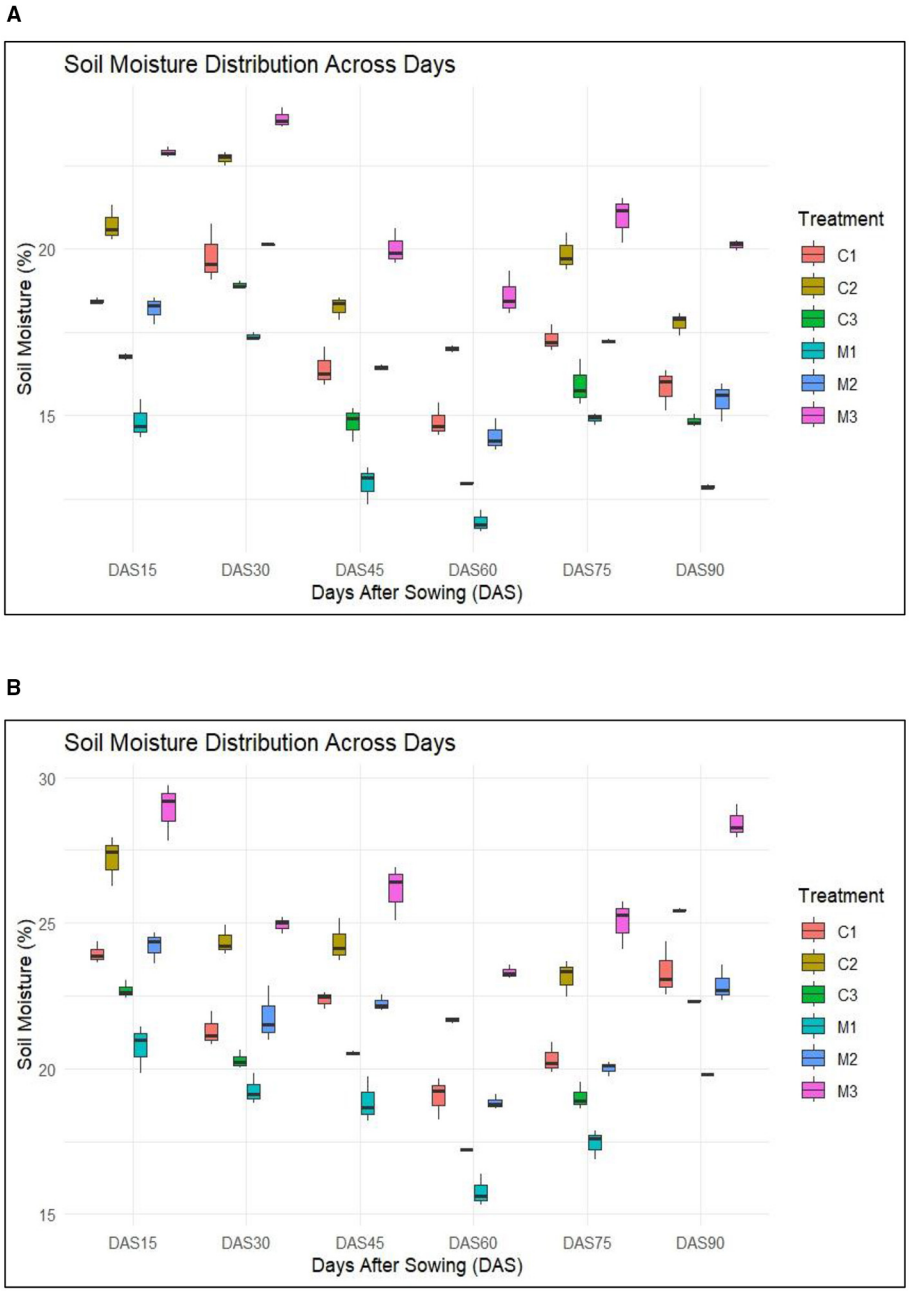


FIGURE 3
(A) Soil moisture content at 0–15 cm soil depths as influenced by tillage practices by using ggplot. **(B)** Soil moisture content at 15–30 cm soil depths as influenced by tillage practices by using ggplot.

system and better adaptation to dryland conditions. Green gram (C_1), on the other hand, had the shortest plants with the lowest other growth components, indicating a lower capacity to develop canopy under moisture-limited conditions (Figure 4). Ghosh et al. (2010) and Tripathi et al. (2007) described that compartmental bunding enhances soil moisture availability, promoting better vegetative growth. Ardeshna et al. (2021) noted that Sesamum performs well under conserved moisture conditions due to its drought resilience and efficient resource utilization. The relatively poor growth of green gram under dry conditions is supported by Sharma et al. (2019), who observed that pulses like green gram are more sensitive to moisture stress and show limited canopy expansion (Dhangada et al., 2017; Komatsuzaki and Ohta, 2007). Thus, the current study confirms that both conservation tillage practices and crop type significantly influence plant growth under dryland conditions.

3.4 Grain yield and efficiency indices

The heatmap revealed clear variability among treatments across multiple parameters. Compartmental bunding (M_3) and field bean (C_2) consistently exhibited higher normalized values for yield, and efficiency indices (Figure 5). The greater performance

TABLE 3 Soil temperature (°C) at different soil depth as influenced by tillage operation on crops.

Treatments	At flowering stage		At harvesting stage	
	5cm	10 cm	5cm	10 cm
Main Plot: moisture conservation practices				
M_1: conventional tillage	28.1 ± 2a	27.5 ± 0.7a	32.5 ± 0.3a	31.4 ± 0.8a
M_2: duck foot cultivator	26.7 ± 0.7b	26.4 ± 1.5b	31.6 ± 0.9ab	30.4 ± 0ab
M_3: compartmental bund	25.9 ± 0.9bc	25.7 ± 1.5bc	30.6 ± 1.3bc	29.8 ± 0.4bc
S.Em±	0.31	0.21	0.31	0.31
CD ($p = 0.05$)	1.22*	0.81*	1.23*	1.21*
Sub plot: crops				
C_1: green gram	27.6 ± 0.1a	27.6 ± 1a	32.6 ± 0.6a	31.7 ± 0.3a
C_2: field bean	26.2 ± 0.6bc	26 ± 1.8abc	30.3 ± 2bc	29.4 ± 0.2bc
C_3: sesamum	26.8 ± 0.6ab	26 ± 1.4ab	31.8 ± 1.7ab	30.6 ± 1.5ab
S.Em±	0.34	0.45	0.57	0.55
CD ($p = 0.05$)	1.04*	1.39*	1.76*	1.71*
Interaction (M×C): NS				

Treatment details are given under materials and methods;
S-Significant ($p = 0.01$*; $p = 0.05$**); NS, Non significant.
The values represent the averages (±SE) of independent replicates, significantly different at $P \leq 0.05$, according to Duncan's Multiple Range test.

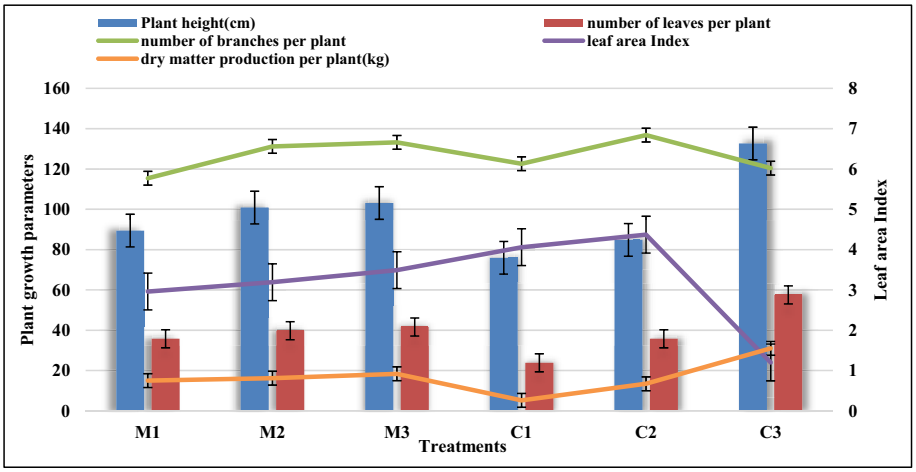


FIGURE 4
Growth attributes influenced by cultivation methods.

of compartmental bunding may be attributed to its ability to reduce runoff and enhance in-situ rainwater harvesting, thereby refining soil moisture accessibility during acute crop progress stages. Comparable findings were stated by Jat et al. (2014) and Lal (2015), who emphasized that conservation practices such as bunding improve soil moisture withholding and water productivity in rainfed systems. Overall, the results suggest that compartmental bunding statistically enhances productivity (894 and 1,276 kg ha^{-1}, seed and greengram equivalent yield, respectively), rainwater use efficiency (2.6), and SYI (0.91) under rainfed pre-monsoon conditions. The clustering pattern suggests that conservation practices are strongly associated with improved system performance due to enhanced soil moisture conservation and reduced degradation (Gathala et al., 2015). The duck foot cultivator (M_2) showed moderate performance with improved yield over conventional tillage due to better soil loosening and infiltration, though its effectiveness was lower than compartmental bunding due to the absence of runoff-controlling structures, as also noted by. Conventional tillage (M_1) recorded the lowest productivity and RWUE, likely due to higher runoff losses and poor soil moisture conservation, which is dependable with findings reported by Ranjan et al. (2006). Whereas among crops, field bean found to be notably superior in yield (1,099 kg ha^{-1}), green gram equivalent yield (GGEY) (1,320 kg ha^{-1}), rain water use efficiency (RWUE) (2.8 kg ha-mm^{-1}) and sustainability yield index (SYI) (0.69). In contrast, sesamum (C_3) showed lower values for most parameters, particularly RWUE and yield index, indicating lower sustainability under the experimental conditions (Figure 5). This is

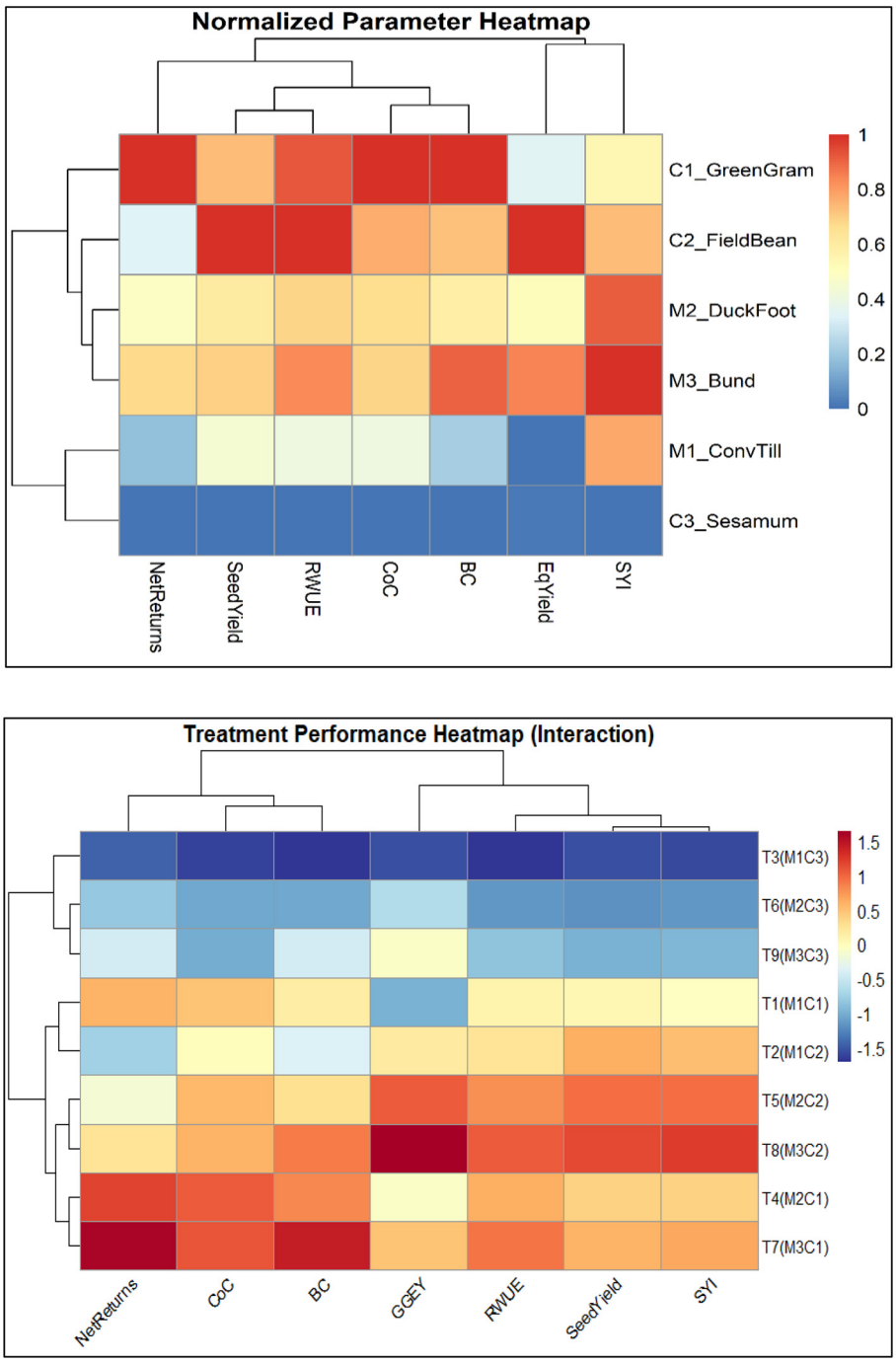


FIGURE 5
Effect of moisture conservation practices, different crops and their combinations on yield and economics in pre-monsoon season (Pooled of 3 years 2022–2024) for interaction and individual treatments. The results are illustrated using heat map in R program for better visualization. Values are standardized as Z-scores, where warm colors indicate above-average performance and cool colors indicate below-average performance.

possibly due to its sensitivity to moisture stress during early growth stages, similar to findings described by Adil (2022) and Wani et al. (2007).

Treatments T_8 ($M3C_2$), T_7 (M3C1), and T_5 (M_2C_2) clustered together and exhibited consistently higher seed yield, Greengram equivalent yield, net returns, B:C ratio, Rain WUE, and SYI, whereas T_3 (M1C3) and T_6 (M_2C3) formed a contrasting cluster with lower performance across most traits. The positive co-clustering of productivity, profitability, water-use efficiency, and yield stability parameters suggests that the superior treatments simultaneously enhanced agronomic and economic performance, with T_8 emerging as the most favorable treatment combination. These outcomes align with recent findings emphasizing the importance of integrating conservation agriculture with legume-based cropping systems for sustainable intensification practices in improving productivity and profitability (Jat et al., 2020;

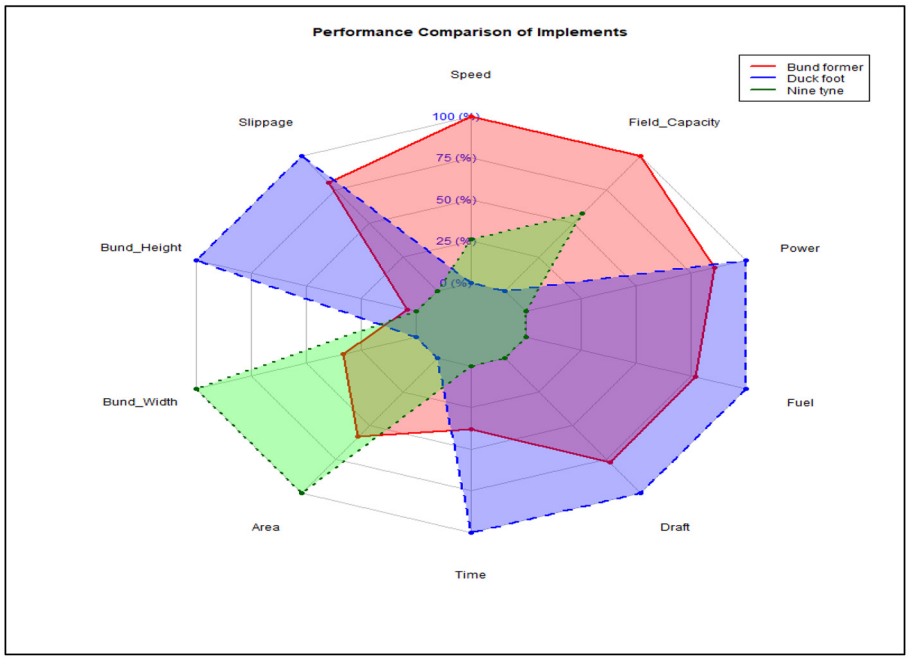


FIGURE 6
Performance evaluation of different tillage implements (Pooled of 3 years 2022–2024). The results are illustrated using radar plot in R program for better comparative visualization.

TABLE 4 Scaled data of the Performance evaluation of different tillage implements (Pooled of 3 years 2022–2024).

Tillage	Speed	Slippage	Bund Height	Bund Width	Area	Time	Draft	Fuel	Power	Field Capacity
Bund former	1.117	0.380	−0.543	−0.218	0.111	−0.157	0.347	0.344	0.442	0.949
Duck foot	−0.812	0.754	1.154	−0.873	−1.051	1.069	0.781	0.783	0.703	−1.044
Nine tyne	−0.305	−1.134	−0.611	1.091	0.940	−0.912	−1.127	−1.126	−1.145	0.095

Duarte et al., 2025). Heatmap analysis based on standardized trait values revealed clear treatment differentiation.

3.5 Economic analysis

According to the economic research, moisture conservation methods and crop choice had a substantial impact on profitability during the pre-monsoon season. Compartmental bunding (M_3) achieved the highest net yields (52,512 ha^{-1}) and B:C ratio (2.8), followed by duck foot cultivator (M_2) and conventional tillage (M_1). The higher performance of compartmental bunding can be related to its ability to limit runoff and improve in-situ moisture conservation, hence increasing soil moisture availability and crop yield under rainfed circumstances (Figure 5).

Similar conclusions have been told by Rattan (2001) and Singh et al. (2015), emphasized that moisture conservation practices significantly enhance yield and farm profitability in dryland agriculture. While green gram (C_1) recorded statistically highest net returns (61,194 ha^{-1}) and benefit cost ratio (2.9), likely due to its short duration, lower cost of cultivation, and higher market demand related to field bean and sesamum. The interaction effect revealed that compartmental bunding with green gram produced the highest net returns and B:C ratio (50,630 ha^{-1} and 2.8), indicating the complementary benefits of effective moisture conservation and efficient moisture utilization by pulse crops (Figure 5). These results are consistent with earlier observations that integrating moisture conservation practices with pulse-based cropping systems improves productivity and economic returns under rainfed conditions (Kumar et al., 2023). Overall, the findings suggest that compartmental bunding combined with green gram or field bean is a more profitable and sustainable option for rainfed pre-monsoon cropping systems.

3.6 Performance evaluation of different tillage implements

The radar chart provided a comprehensive comparison of all performance parameters simultaneously. It clearly indicated that the nine-tyne spring tooth cultivator exhibited relatively balanced performance across most parameters (Figure 6). The duck foot cultivator exhibited superior bund height (29.8 cm),

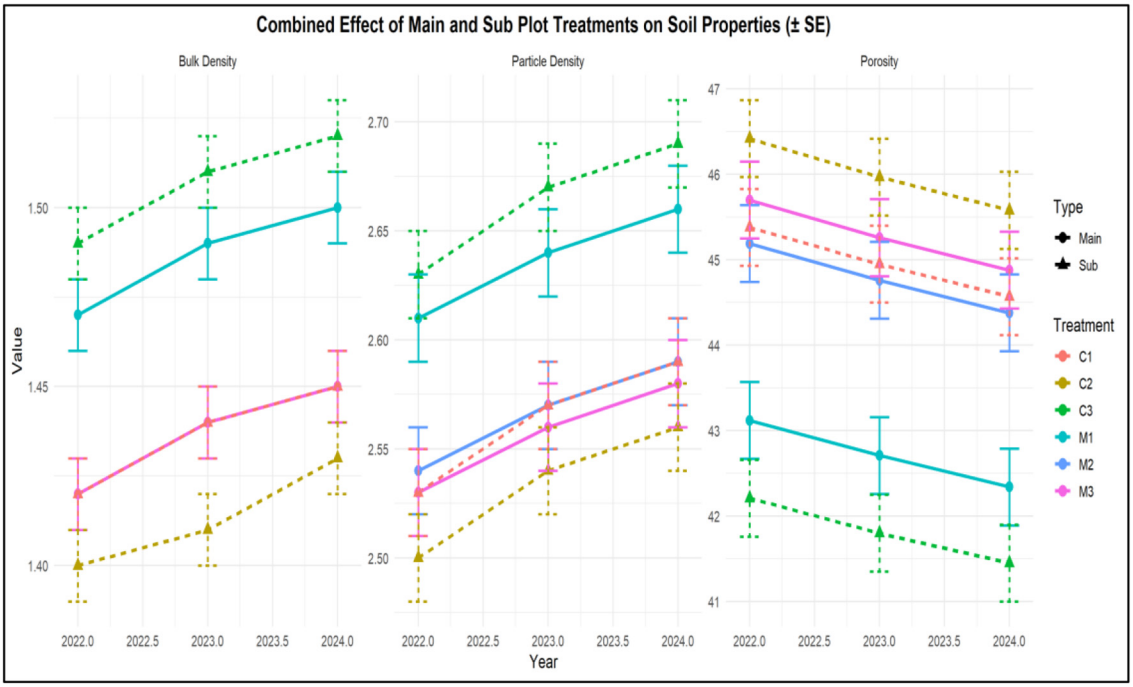


FIGURE 7
Effect of different tillage implements on soil physical parameters (Pooled of 3 years 2022–2024). The inclusion of standard error bars indicates that the variation among treatments is consistent and statistically reliable. Differences exceeding the critical difference (CD at 5%) can be considered significant, particularly in porosity and bulk density, confirming the effectiveness of conservation practices and crop selection.

which bring into line with findings of Bainer et al. (1956), stated that aggressive soil-engaging implements produce higher soil displacement. However, this increased bund formation resulted in higher draft force (535.7 kgf), fuel consumption (5.9 lph), and power requirement (7.9 hp), consistent with the relationship between soil resistance and energy demand (ASABE, 2011) Recent studies confirm that tillage depth and soil engagement significantly increase draft force and energy consumption (Kim et al., 2020). The increased energy demand observed in the duck foot cultivator is therefore expected. In contrast, the compartmental bund former demonstrated moderate performance across most parameters but did not excel in any particular aspect. The nine tyne cultivator showed lower draft force (276.3 kgf), reduced fuel consumption (4.8 lph), and lower power requirement (4.1 hp), indicating better energy efficiency. Similar trends were reported by Srivastava et al. (2006), detected that lighter tillage implements require less tractive effort and fuel (Figure 6). Additionally, it achieved the highest area coverage (0.82 ha hr^{-1}) and lowest time requirement (1.22 h ha^{-1}), making it operationally efficient. Modern research also highlights that optimizing tillage parameters can maintain high power output while reducing fuel consumption (Han et al., 2023; Table 4).

3.7 Soil physical properties

The combined graphical analysis depicting mean ± SE clearly illustrates the influence of main plot (moisture conservation practices) and sub-plot (cropping systems) treatments on soil physical properties over the years (2022–2024; Figure 7). Bulk density exhibited a gradual increase across years under all treatments, indicating progressive soil consolidation; however, conservation practices such as duck foot cultivator (M_2) and compartmental bunding (M_3) maintained relatively lower values compared to conventional tillage (M_1), reflecting improved soil structure under reduced disturbance (Blanco-Canqui and Ruis, 2018; Blevins et al., 2018). Among cropping systems, sesamum (C_3) consistently recorded higher bulk density, whereas field bean (C_2) showed lower values, suggesting improved soil structure under leguminous crops due to greater root activity and organic matter addition (Duarte et al., 2025). The lower bulk density under leguminous crops can be attributed to enhanced root activity and organic matter addition, which improves soil aggregation and reduces compaction. Particle density exhibited a slight increasing trend with marginal variation among treatments; however, conventional tillage tended to show marginally higher values. Among cropping systems, sesamum recorded a higher particle density compared to legumes. This may be due to lower biomass return and organic matter contribution in non-leguminous crops, resulting in comparatively denser soil particles (Friedrich and Kassam, 2011). Porosity exhibited a declining trend over the years, reflecting gradual soil compaction. However, conservation practices (M_2 and M_3) maintained higher porosity compared to conventional tillage (Figure 7). Legume-based systems (sub-plot treatments), particularly field bean (C_2), recorded the highest porosity, followed by green gram (C_1), while sesamum (C_3) showed the lowest, due to improved aggregation and pore continuity. The higher porosity under legumes can be attributed to improved soil structure through root proliferation and microbial activity, which enhance soil aggregation and

pore space distribution (Gathala et al., 2015; Teng et al., 2024). The conservation tillage practices combined with leguminous cropping systems improve soil physical properties by reducing bulk density and enhancing porosity, thereby contributing to sustainable soil health management (Blanco-Canqui and Ruis, 2018; Cárceles Rodríguez et al., 2022).

4 Conclusion

Compartmental bunding (M_3) significantly improved soil moisture conservation, crop productivity, and resource-use efficiency under pre-monsoon Alfisol conditions. Compared with conventional tillage (M_1), compartmental bunding increased seed yield, green gram equivalent yield, and rainwater use efficiency by 23.0%, 23.5%, and 24.6%, respectively, while enhancing net returns and benefit-cost ratio by 34.2% and 15.7%. Among the crop combinations, compartmental bunding combined with field bean (*Dolichos lablab* L.) var. HA-4 (T_8) recorded the highest seed yield (997 kg ha^{-1}), green gram equivalent yield (1,298 kg ha^{-1}), rainwater use efficiency (2.7 kg ha-mm^{-1}), and sustainable yield index (0.8). The treatment also maintained greater soil moisture and lower soil temperature throughout the crop growth period. Therefore, the combination of compartmental bunding and field bean (HA-4) can be recommended as an effective moisture conservation strategy for enhancing productivity, water-use efficiency, and sustainability in pre-monsoon dryland Alfisols.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding authors.

Author contributions

HL: Conceptualization, Investigation, Methodology, Resources, Supervision, Writing – original draft, Writing – review & editing. M: Investigation, Supervision, Writing – review & editing. NU: Data curation, Formal analysis, Visualization, Writing – review & editing. BV: Writing – review & editing. KD: Writing – review & editing. KG: Writing – review & editing. MM: Writing – review & editing. MS: Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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